01 BASIC LINUX COMMANDS

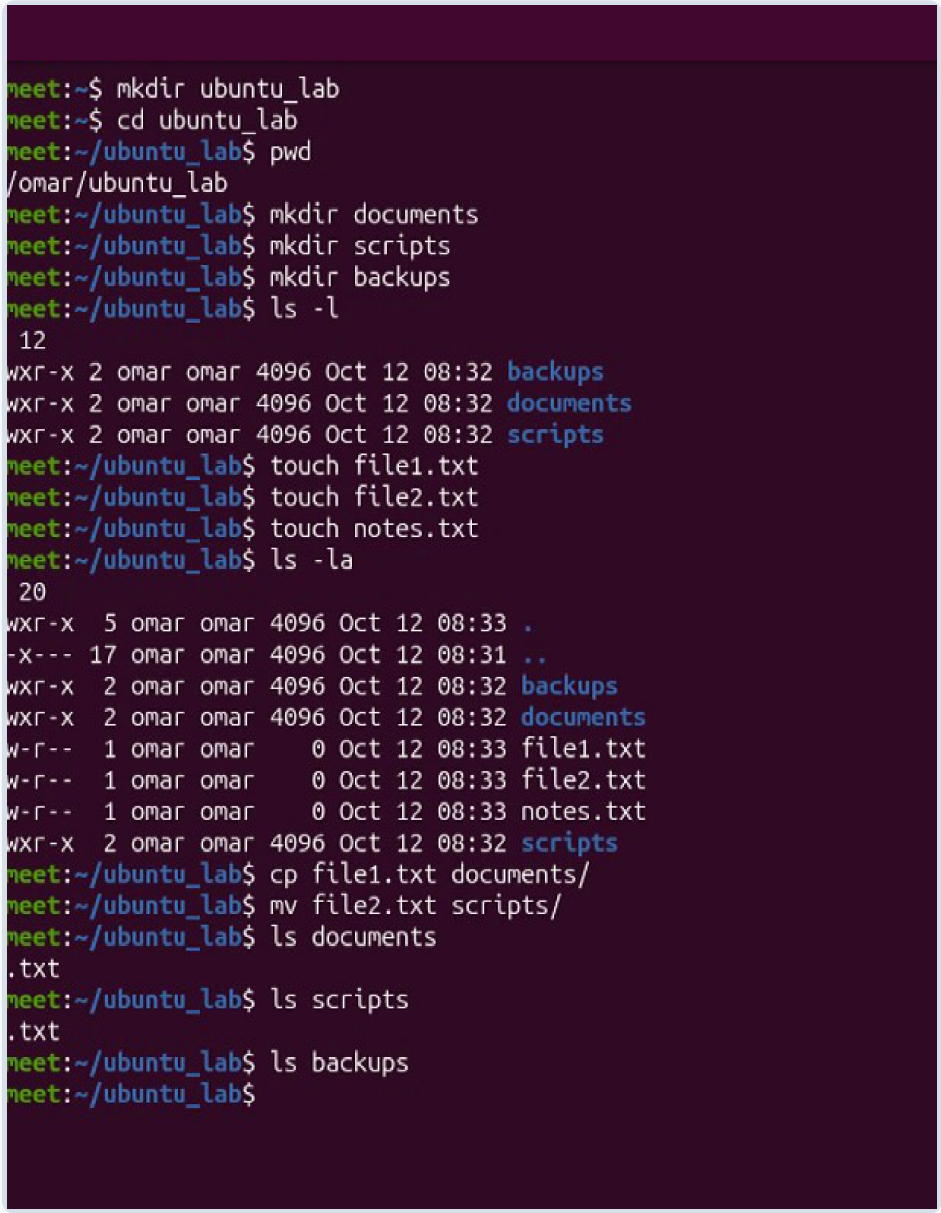
System identity, navigation and file listing

```
omar@meet:~/Desktop$ whoami
omar
omar@meet:~/Desktop$ hostname
meet
omar@meet:~/Desktop$ pwd
/home/omar/Desktop
omar@meet:~/Desktop$ date
Mon Oct 12 08:28:44 AM EEST 2026
omar@meet:~/Desktop$ uname -a
Linux meet 7.0.0-28-generic #28-Ubuntu SMP PREEMPT_DYNAMIC Sun Jun 21 01:01:36 UTC 2026 x86_64 GNU/Linux
omar@meet:~/Desktop$ ls
openssl.cnf  server.crt  server.csr  server.key
omar@meet:~/Desktop$ ls -l
total 16
-rw-rw-r-- 1 omar omar  280 Sep 21 09:19 openssl.cnf
-rw-rw-r-- 1 omar omar 1330 Sep 21 09:19 server.crt
-rw-rw-r-- 1 omar omar 1163 Sep 21 09:13 server.csr
-rw------- 1 omar omar 1704 Sep 21 09:08 server.key
omar@meet:~/Desktop$ ls -a
.  ..  openssl.cnf  server.crt  server.csr  server.key
omar@meet:~/Desktop$ ls -la
total 24
drwxr-xr-x  2 omar omar 4096 Sep 21 09:19 .
drwxr-x--- 16 omar omar 4096 Oct  7 14:20 ..
-rw-rw-r--  1 omar omar  280 Sep 21 09:19 openssl.cnf
-rw-rw-r--  1 omar omar 1330 Sep 21 09:19 server.crt
-rw-rw-r--  1 omar omar 1163 Sep 21 09:13 server.csr
-rw-------  1 omar omar 1704 Sep 21 09:08 server.key
omar@meet:~/Desktop$ cd /home
omar@meet:/home$ pwd
/home
omar@meet:/home$ ls
omar
omar@meet:/home$ cd ~
omar@meet:~$ pwd
/home/omar
```

Original terminal output - unmodified

02 FILES AND DIRECTORIES

Creating and organizing files and folders



```
meet:~$ mkdir ubuntu_lab
meet:~$ cd ubuntu_lab
meet:~/ubuntu_lab$ pwd
/omar/ubuntu_lab
meet:~/ubuntu_lab$ mkdir documents
meet:~/ubuntu_lab$ mkdir scripts
meet:~/ubuntu_lab$ mkdir backups
meet:~/ubuntu_lab$ ls -l
 12
wxr-x 2 omar omar 4096 Oct 12 08:32 backups
wxr-x 2 omar omar 4096 Oct 12 08:32 documents
wxr-x 2 omar omar 4096 Oct 12 08:32 scripts
meet:~/ubuntu_lab$ touch file1.txt
meet:~/ubuntu_lab$ touch file2.txt
meet:~/ubuntu_lab$ touch notes.txt
meet:~/ubuntu_lab$ ls -la
 20
wxr-x  5 omar omar 4096 Oct 12 08:33 .
-x--- 17 omar omar 4096 Oct 12 08:31 ..
wxr-x  2 omar omar 4096 Oct 12 08:32 backups
wxr-x  2 omar omar 4096 Oct 12 08:32 documents
w-r--  1 omar omar    0 Oct 12 08:33 file1.txt
w-r--  1 omar omar    0 Oct 12 08:33 file2.txt
w-r--  1 omar omar    0 Oct 12 08:33 notes.txt
wxr-x  2 omar omar 4096 Oct 12 08:32 scripts
meet:~/ubuntu_lab$ cp file1.txt documents/
meet:~/ubuntu_lab$ mv file2.txt scripts/
meet:~/ubuntu_lab$ ls documents
.txt
meet:~/ubuntu_lab$ ls scripts
.txt
meet:~/ubuntu_lab$ ls backups
meet:~/ubuntu_lab$
```

Original terminal output - unmodified

03 FILE CONTENT MANAGEMENT

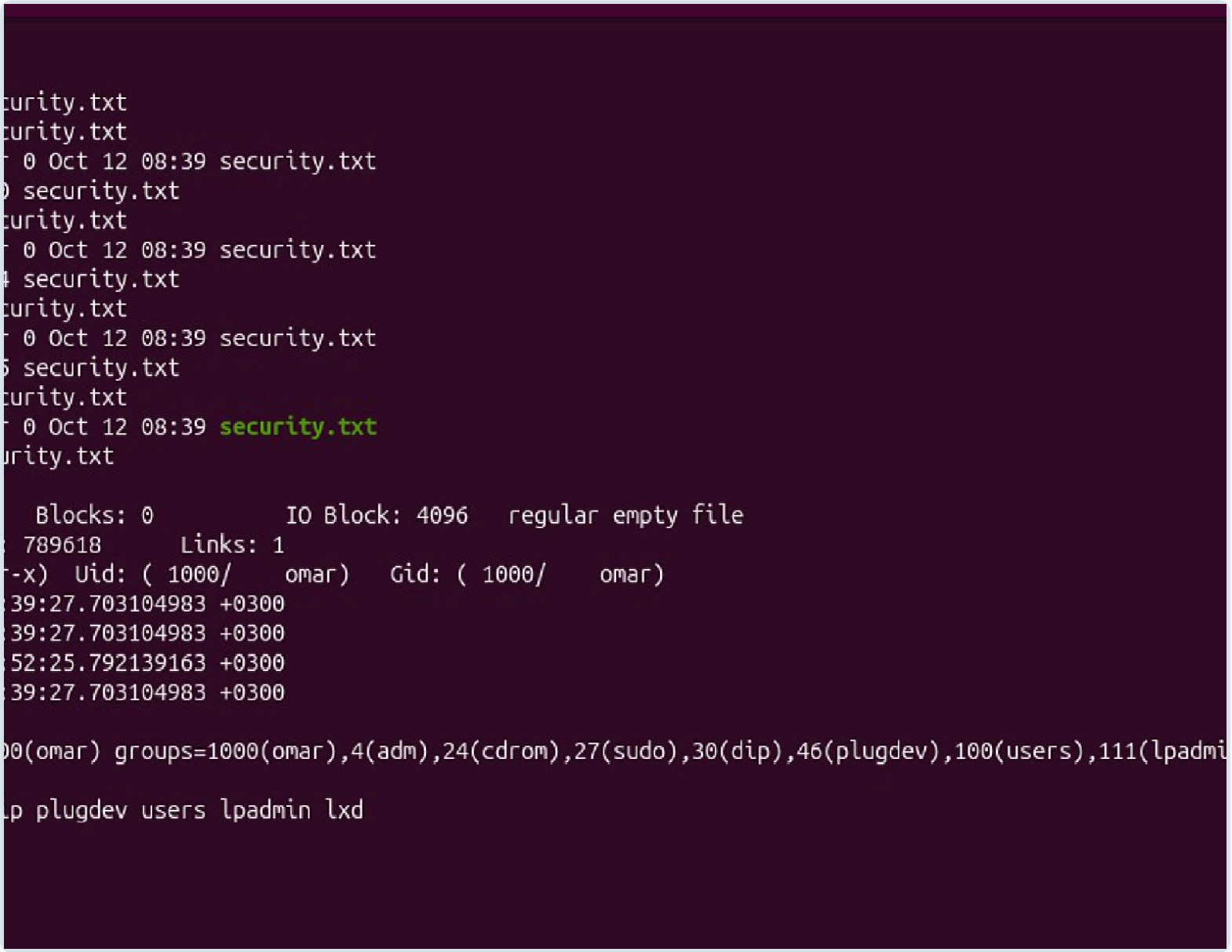
Writing, reading and copying text files

```
omar@meet:~$ cat notes.txt
Hello Ubuntu
omar@meet:~$ echo "Linux Administration Lab" >> notes.txt
omar@meet:~$ echo "Student Practical Work" >> notes.txt
omar@meet:~$ cat notes.txt
Hello Ubuntu
Linux Administration Lab
Student Practical Work
omar@meet:~$ wc -l notes.txt
3 notes.txt
omar@meet:~$ wc -w notes.txt
8 notes.txt
omar@meet:~$ head notes.txt
Hello Ubuntu
Linux Administration Lab
Student Practical Work
omar@meet:~$ tail notes.txt
Hello Ubuntu
Linux Administration Lab
Student Practical Work
omar@meet:~$ cp notes.txt backup_notes.txt
omar@meet:~$ ls -l
total 52
drwxr-xr-x 2 omar omar 4096 Sep 21 09:19 Desktop
drwxr-xr-x 2 omar omar 4096 Aug  3 10:35 Documents
drwxr-xr-x 2 omar omar 4096 Aug  3 10:35 Downloads
drwxr-xr-x 2 omar omar 4096 Aug  3 10:35 Music
drwxr-xr-x 2 omar omar 4096 Aug  3 10:35 Pictures
drwxr-xr-x 2 omar omar 4096 Aug  3 10:35 Public
drwxr-xr-x 2 omar omar 4096 Aug  3 10:35 Templates
drwxr-xr-x 2 omar omar 4096 Aug  3 10:35 Videos
-rw-rw-r-- 1 omar omar   61 Oct 12 08:37 backup_notes.txt
drwxrwxr-x 2 omar omar 4096 Oct  7 14:20 lab
-rw-rw-r-- 1 omar omar   61 Oct 12 08:36 notes.txt
drwx------ 7 omar omar 4096 Aug  4 09:47 snap
```

Original terminal output - unmodified

04 FILE PERMISSIONS

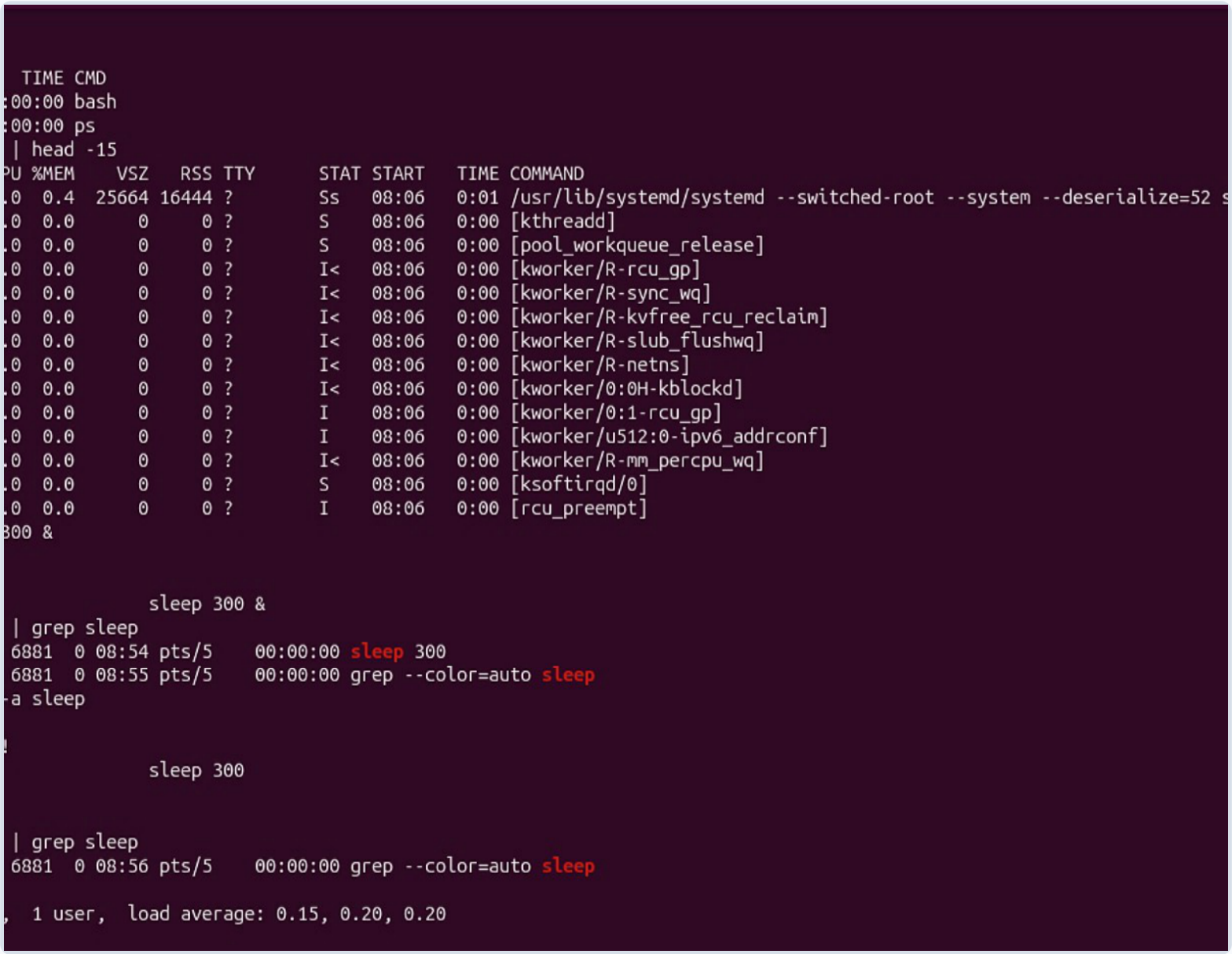
chmod, stat and user groups



Original terminal output - unmodified

05 PROCESS MANAGEMENT

Listing, starting and stopping processes



```
  TIME CMD
:00:00 bash
:00:00 ps
 | head -15
PU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
.0  0.4  25664 16444 ?        Ss   08:06   0:01 /usr/lib/systemd/systemd --switched-root --system --deserialize=52 s
.0  0.0      0     0 ?        S    08:06   0:00 [kthreadd]
.0  0.0      0     0 ?        S    08:06   0:00 [pool_workqueue_release]
.0  0.0      0     0 ?        I<   08:06   0:00 [kworker/R-rcu_gp]
.0  0.0      0     0 ?        I<   08:06   0:00 [kworker/R-sync_wq]
.0  0.0      0     0 ?        I<   08:06   0:00 [kworker/R-kvfree_rcu_reclaim]
.0  0.0      0     0 ?        I<   08:06   0:00 [kworker/R-slub_flushwq]
.0  0.0      0     0 ?        I<   08:06   0:00 [kworker/R-netns]
.0  0.0      0     0 ?        I<   08:06   0:00 [kworker/0:0H-kblockd]
.0  0.0      0     0 ?        I    08:06   0:00 [kworker/0:1-rcu_gp]
.0  0.0      0     0 ?        I    08:06   0:00 [kworker/u512:0-ipv6_addrconf]
.0  0.0      0     0 ?        I<   08:06   0:00 [kworker/R-mm_percpu_wq]
.0  0.0      0     0 ?        S    08:06   0:00 [ksoftirqd/0]
.0  0.0      0     0 ?        I    08:06   0:00 [rcu_preempt]
300 &


            sleep 300 &
 | grep sleep
 6881  0 08:54 pts/5    00:00:00 sleep 300
 6881  0 08:55 pts/5    00:00:00 grep --color=auto sleep
-a sleep

1
            sleep 300

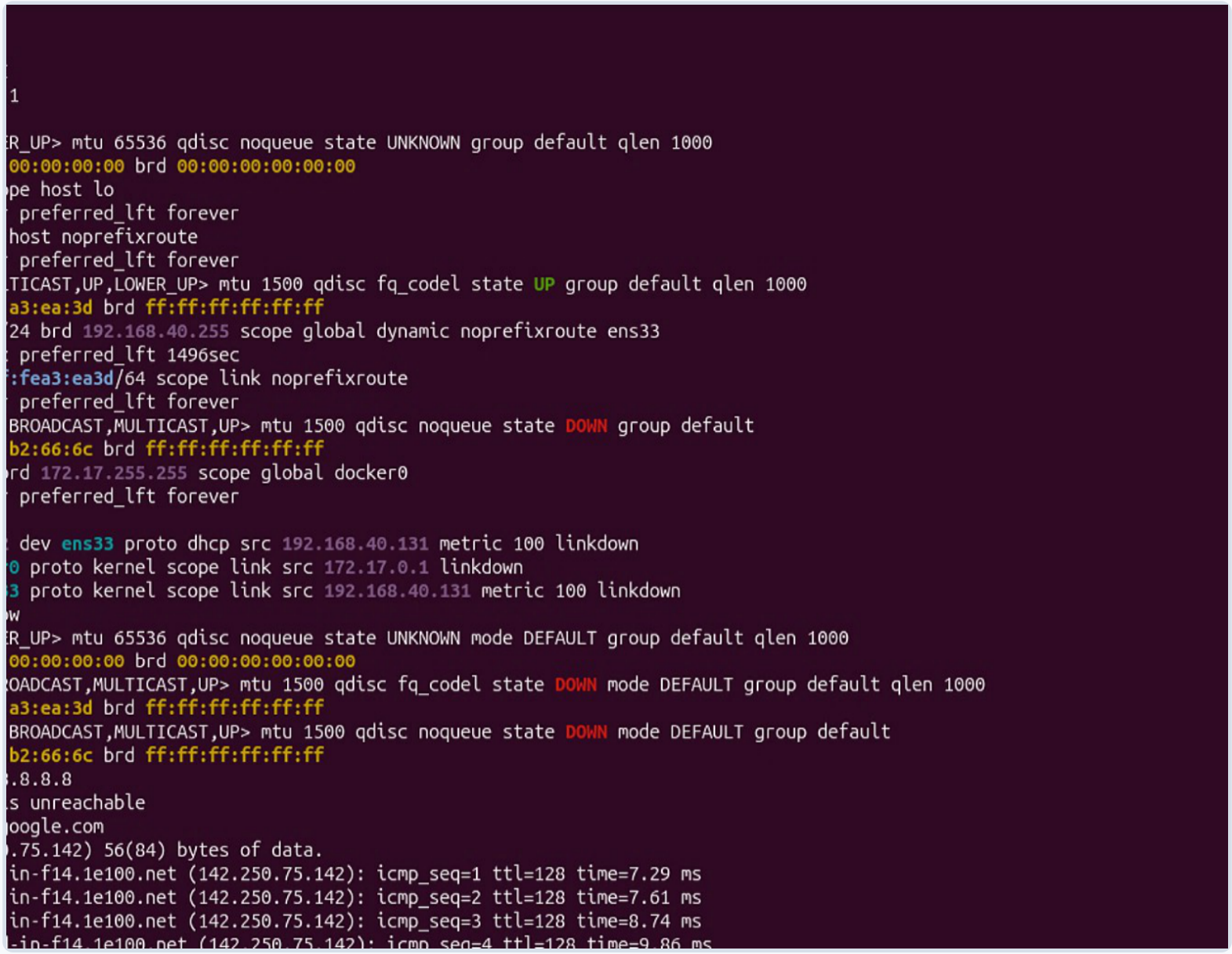

 | grep sleep
 6881  0 08:56 pts/5    00:00:00 grep --color=auto sleep

,  1 user,  load average: 0.15, 0.20, 0.20
```

Original terminal output - unmodified

06 NETWORK CONFIGURATION

IP interfaces, routing and connectivity



```
1

R_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
00:00:00:00 brd 00:00:00:00:00:00
pe host lo
 preferred_lft forever
host noprefixroute
 preferred_lft forever
TICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
a3:ea:3d brd ff:ff:ff:ff:ff:ff
24 brd 192.168.40.255 scope global dynamic noprefixroute ens33
 preferred_lft 1496sec
:fea3:ea3d/64 scope link noprefixroute
 preferred_lft forever
BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
b2:66:6c brd ff:ff:ff:ff:ff:ff
rd 172.17.255.255 scope global docker0
 preferred_lft forever

 dev ens33 proto dhcp src 192.168.40.131 metric 100 linkdown
0 proto kernel scope link src 172.17.0.1 linkdown
3 proto kernel scope link src 192.168.40.131 metric 100 linkdown
w
R_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
00:00:00:00 brd 00:00:00:00:00:00
OADCAST,MULTICAST,UP> mtu 1500 qdisc fq_codel state DOWN mode DEFAULT group default qlen 1000
a3:ea:3d brd ff:ff:ff:ff:ff:ff
BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN mode DEFAULT group default
b2:66:6c brd ff:ff:ff:ff:ff:ff
.8.8.8
s unreachable
oogle.com
.75.142) 56(84) bytes of data.
in-f14.1e100.net (142.250.75.142): icmp_seq=1 ttl=128 time=7.29 ms
in-f14.1e100.net (142.250.75.142): icmp_seq=2 ttl=128 time=7.61 ms
in-f14.1e100.net (142.250.75.142): icmp_seq=3 ttl=128 time=8.74 ms
-in-f14.1e100.net (142.250.75.142): icmp_seq=4 ttl=128 time=9.86 ms
```

Original terminal output - unmodified

07 REDIRECTION AND PIPES

Output, errors, grep and pipelines

```
d

ho "Linux Practical Exam" > output.txt
t output.txt

ho "Ubuntu Commands" >> output.txt
t output.txt


 -l > files.txt
t files.txt

   0 Oct 12 09:02 files.txt
 280 Sep 21 09:19 openssl.cnf
  37 Oct 12 09:00 output.txt
1330 Sep 21 09:19 server.crt
1163 Sep 21 09:13 server.csr
1704 Sep 21 09:08 server.key
 /nonexistent_directory 2> errors.txt
t errors.txt
existent_directory': No such file or directory
t output.txt | grep Ubuntu

t output.txt | wc -l

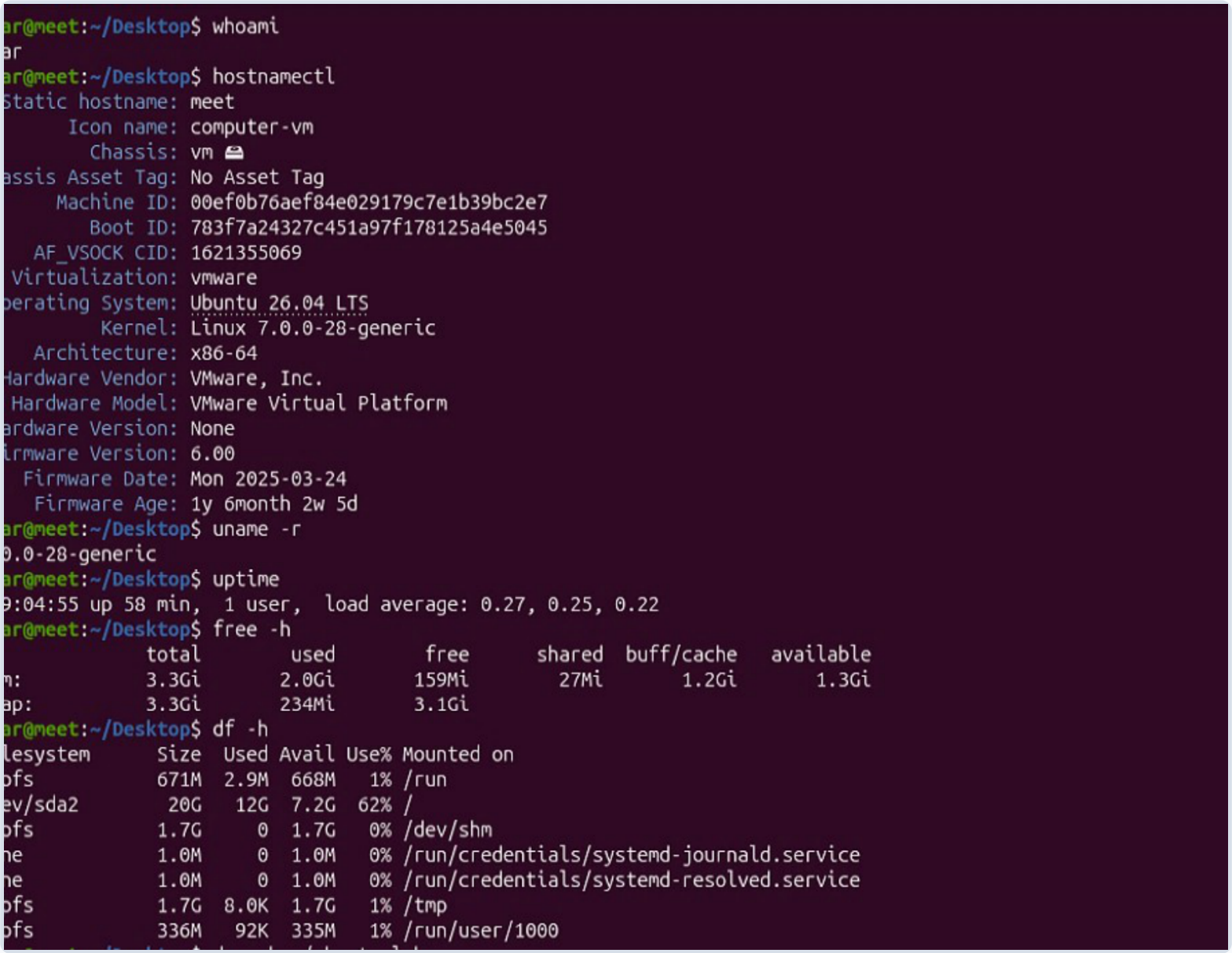
 -l | head -5

  70 Oct 12 09:02 errors.txt
 321 Oct 12 09:02 files.txt
 280 Sep 21 09:19 openssl.cnf
  37 Oct 12 09:00 output.txt
```

Original terminal output - unmodified

08 SYSTEM MONITORING - PART 1

System identity, memory and disk usage



```
ar@meet:~/Desktop$ whoami
ar
ar@meet:~/Desktop$ hostnamectl
Static hostname: meet
      Icon name: computer-vm
        Chassis: vm
assis Asset Tag: No Asset Tag
     Machine ID: 00ef0b76aef84e029179c7e1b39bc2e7
        Boot ID: 783f7a24327c451a97f178125a4e5045
   AF_VSOCK CID: 1621355069
 Virtualization: vmware
perating System: Ubuntu 26.04 LTS
         Kernel: Linux 7.0.0-28-generic
   Architecture: x86-64
Hardware Vendor: VMware, Inc.
 Hardware Model: VMware Virtual Platform
ardware Version: None
irmware Version: 6.00
  Firmware Date: Mon 2025-03-24
   Firmware Age: 1y 6month 2w 5d
ar@meet:~/Desktop$ uname -r
0.0-28-generic
ar@meet:~/Desktop$ uptime
9:04:55 up 58 min,  1 user,  load average: 0.27, 0.25, 0.22
ar@meet:~/Desktop$ free -h
               total        used        free      shared  buff/cache   available
m:            3.3Gi       2.0Gi       159Mi        27Mi       1.2Gi       1.3Gi
ap:           3.3Gi       234Mi       3.1Gi
ar@meet:~/Desktop$ df -h
lesystem      Size  Used Avail Use% Mounted on
ofs           671M  2.9M  668M   1% /run
ev/sda2        20G   12G  7.2G  62% /
ofs           1.7G     0  1.7G   0% /dev/shm
he            1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
he            1.0M     0  1.0M   0% /run/credentials/systemd-resolved.service
ofs           1.7G  8.0K  1.7G   1% /tmp
ofs           336M   92K  335M   1% /run/user/1000
```

Original terminal output - unmodified

09 SYSTEM MONITORING - PART 2

Storage devices and CPU information

```
Desktop$ du -sh ~/ubuntu_lab
e/omar/ubuntu_lab
Desktop$ lsblk
IN RM   SIZE RO TYPE MOUNTPOINTS
    0     4K  1 loop /snap/bare/5
    0  66.8M  1 loop /snap/core24/2124
    0    20M  1 loop /snap/desktop-security-center/151
    0  66.8M  1 loop /snap/core24/1643
    0  14.2M  1 loop /snap/desktop-security-center/206
    0 606.1M  1 loop /snap/gnome-46-2404/153
    0  16.5M  1 loop /snap/firmware-updater/226
    0 273.7M  1 loop /snap/firefox/8107
    0   395M  1 loop /snap/mesa-2404/1165
    0  18.8M  1 loop /snap/prompting-client/222
0   0   580K  1 loop /snap/snapd-desktop-integration/361
1   0  91.7M  1 loop /snap/gtk-common-themes/1535
2   0  15.7M  1 loop /snap/snap-store/1367
3   0  18.4M  1 loop /snap/prompting-client/228
4   0  49.3M  1 loop /snap/snapd/26865
    0    20G  0 disk
    0     1M  0 part
    0    20G  0 part /
    1  1024M  0 rom
Desktop$ lscpu | head -20
:                         x86_64
s):                       32-bit, 64-bit
s:                        45 bits physical, 48 bits virtual
                          Little Endian
                          2
s) list:                  0,1
                          GenuineIntel
                          12th Gen Intel(R) Core(TM) i7-12700K
                          6
                          151
r core:                   1
socket:                   1
                          2
                          2
```

Ubuntu Administration - Automation Appendix

Student: Omar Shuqo | **Topic:** Bash scripting and scheduled execution

Purpose

Automate related Ubuntu administration commands with a reusable Bash script. This appendix documents the proposed implementation and verification steps. It does **not** claim the script or scheduler was executed; the student should run the commands and attach genuine terminal evidence if required.

1. Create the automation script

Open Ubuntu Terminal and create a script:

```
nano ~/automation.sh
```

Paste the following code:

```
#!/usr/bin/env bash
set -e

LAB_DIR="$HOME/automation_lab"
mkdir -p "$LAB_DIR"

printf '\n===== SYSTEM INFORMATION =====\n'
whoami
hostname
date
uname -a

printf '\n===== FILE MANAGEMENT =====\n'
echo "Hello Ubuntu" > "$LAB_DIR/file1.txt"
cat "$LAB_DIR/file1.txt"
ls -la "$LAB_DIR"

printf '\n===== SYSTEM MONITORING =====\n'
free -h
df -h
uptime

printf '\n===== NETWORK INFORMATION =====\n'
hostname -I
ip route

printf '\n===== AUTOMATION COMPLETED =====\n'
```

Save using Ctrl+O, Enter, then Ctrl+X. Make the script executable and run it:

```
chmod +x ~/automation.sh
~/automation.sh
```

2. Capture real execution evidence

Take a screenshot of the Terminal showing the command **~/automation.sh** and its output. Include the final message **AUTOMATION COMPLETED** where possible. Do not substitute example output for an actual execution screenshot.

3. Optional: Schedule automatic execution with cron

Bash scripting runs multiple commands with one invocation. To execute the script automatically on a schedule, configure cron:

```
crontab -e
```

Add the following line to run the script daily at 09:00 (Ubuntu machine local time):

```
0 9 * * * /bin/bash "$HOME/automation.sh" >> "$HOME/automation.log" 2>&1
```

Check the saved schedule:

```
crontab -l
```

To test without waiting for the scheduled time, run:

```
/bin/bash "$HOME/automation.sh" >> "$HOME/automation.log" 2>&1
tail -n 25 "$HOME/automation.log"
```

Cron requires the Ubuntu system to be running at the scheduled time. The log stores both standard output and errors. This is **local automation**, not cloud deployment. If cloud execution was specifically requested, the platform and requirements must be confirmed separately.

Suggested evidence for submission

Attach two additional genuine screenshots if the instructor requests proof: (1) script execution with results, and (2) **crontab -l** plus the log output. These screenshots have not been added because no such screenshots were supplied.

10 AUTOMATION: EXECUTION & VERIFICATION

Bash automation implementation and reproducible verification procedure

Objective: Run system, file, resource, and network commands automatically from a single Bash script, and store the resulting output in a log.

1. Place automation.sh in the Ubuntu home directory, then execute:

```
cd ~
chmod +x automation.sh
./automation.sh
```

2. Inspect generated artifacts:

```
ls -lh ~/automation_lab
cat ~/automation_lab/file1.txt
tail -n 25 ~/automation_lab/automation.log
```

3. Optional scheduled execution with cron:

```
crontab -e
# Add this line (runs daily at 09:00):
0 9 * * * /bin/bash "$HOME/automation.sh"
# Verify the scheduled entry:
crontab -l
```

Evidence required: Take a screenshot showing the actual terminal output after running the script. If cron is configured, capture the output of `crontab -l` as separate evidence.

Submission status: The script and instructions are provided. Automation execution and scheduling are **not independently verified** by the screenshots in this document.